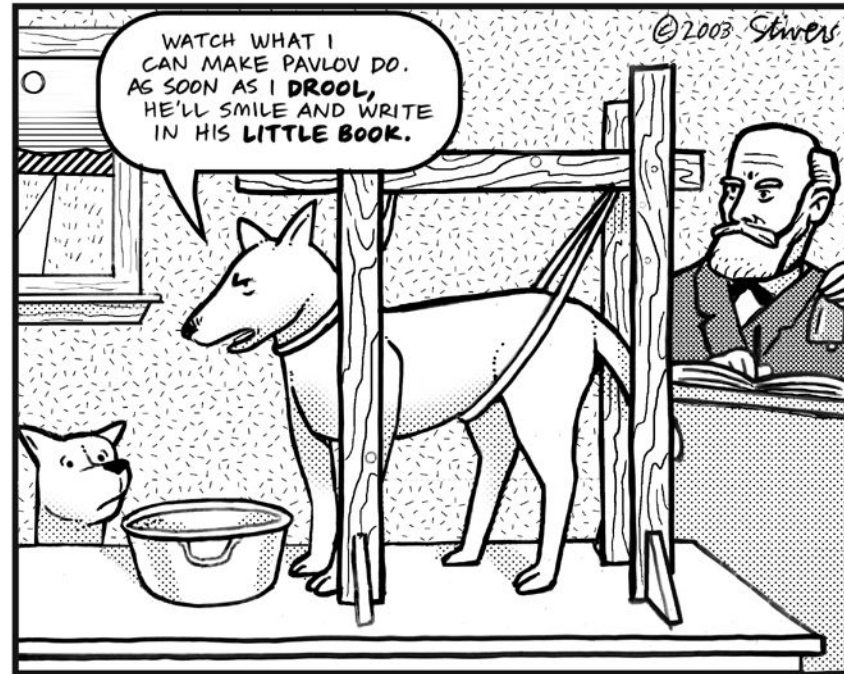


Lecture 8: Conditioning

PSYCH/COGS 140M 2026.04.30



Reminders

- Mid-term evaluations (optional, anonymous) close **tomorrow, 5pm Friday**
- **Quiz #2 next Tuesday**
 - **Online-only** from **12:30pm Tuesday - 6pm Wednesday**
 - **Lockdown browser**

Notes

- Lectures 6-8 cover *non-declarative learning*
 - 6: *Skill learning* - **High-level**: What practices and conditions help us build skills?
 - 7: *Habitization, sensitization, familiarization, and priming* - **More detailed**: What are the component mental and physical processes that comprise skills?
 - 8: *Conditioning* - **Most detailed**: Formally capture the building blocks of procedural learning, building towards a mathematical model.

Outline

- Classical Conditioning Overview
- Rescorla Wagner Model
- Operant Conditioning

“Classic”-al, “Opera”-nt



Outline

- Classical Conditioning Overview
- Rescorla Wagner Model
- Operant Conditioning

Non-associative vs. Associative Learning

● Non-associative learning

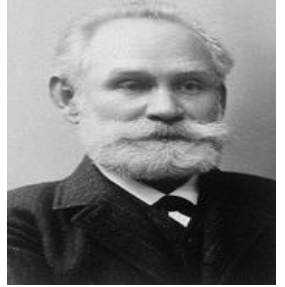
- Change in elicited behavior due to stimulus repetition (learning about a single stimulus)

● Associative learning

- Classical conditioning: learning about associations between stimuli
- Operant conditioning: learning about associations between behaviors and their consequences

Classical Conditioning

- Pioneered by Ivan Pavlov
- Involves pairing a neutral stimulus with one that is either inherently appetitive or aversive



1849 – 1936

Classical Conditioning: Terminology

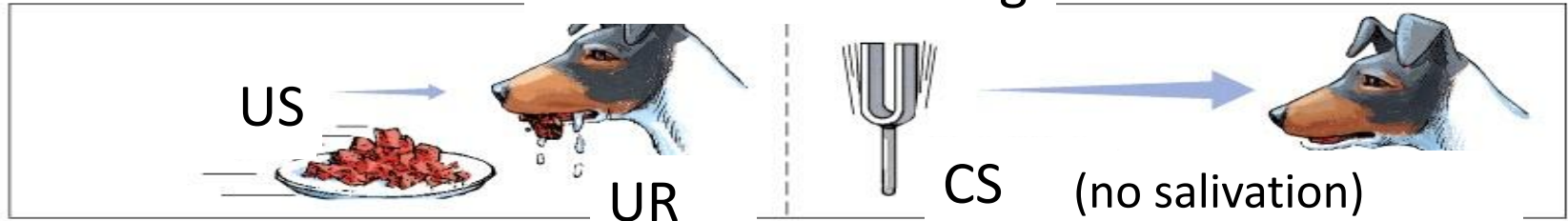
US: unconditioned stimulus

UR: unconditioned response

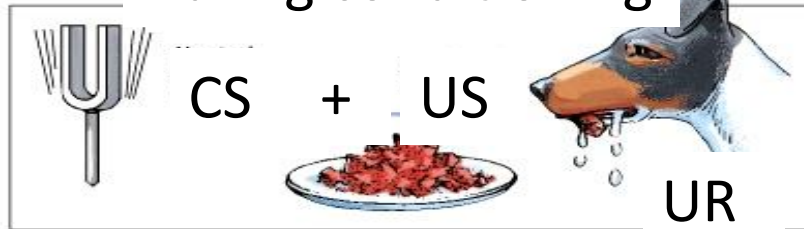
CS: conditioned stimulus

CR: conditioned response

Before conditioning



During conditioning



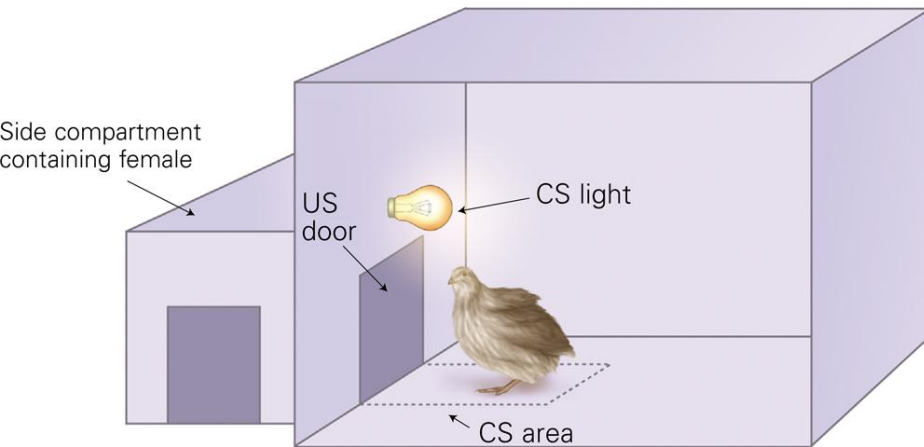
After conditioning



What are some other examples of a CS, US, CR, and UR?

Appetitive Conditioning

- Sustenance and sex are the most powerful appetitive stimuli



- Experiment: light is paired with exposure to female
- (male) Quail learns to approach opening when light goes on
- The CR prepares the animal to receive the US
- The CR and UR are not necessarily the same! (here, **approach** is the CR and **arousal** is the UR)

Aversive Conditioning: Eyeblink

(a) Before training

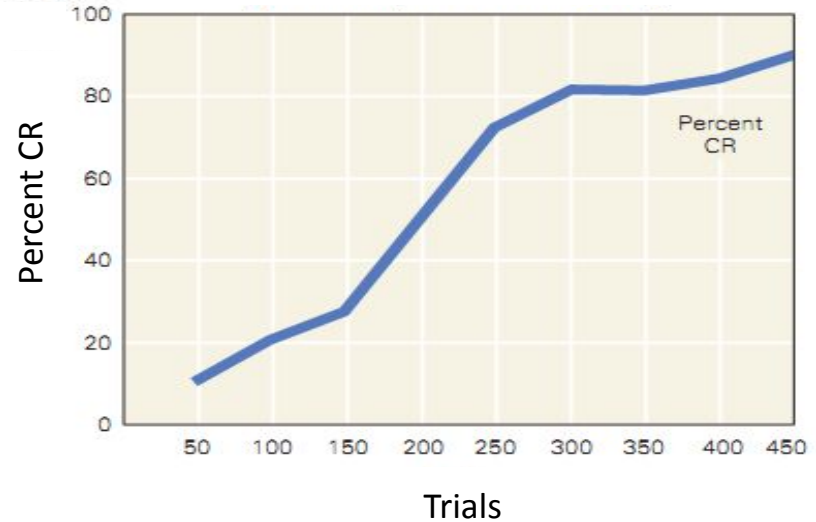


Time



Tone CS:
conditioned
stimulus

Eyeblink CR:
conditioned
response



Aversive Conditioning: Foot Shocks

Fear conditioning



- US = shock
- UR = freezing
- CS = tone
- CR = freezing

What might be the adaptive value of conditioning?

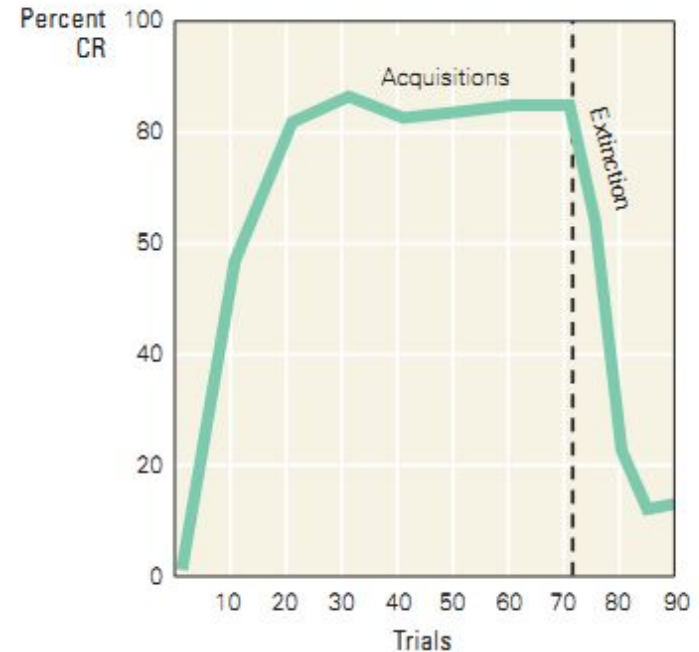
Acquisition and Extinction

- Acquisition

- Acquiring a CR due to repeated CS/US pairing

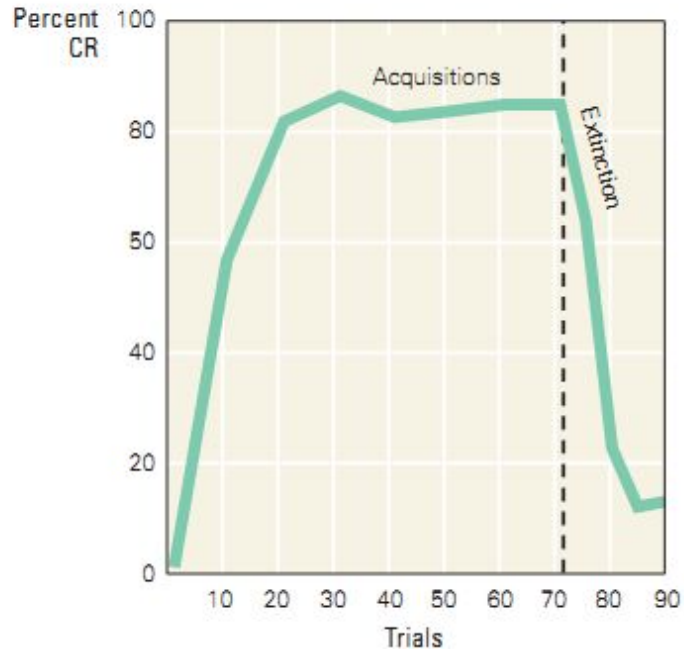
- Extinction

- Losing a CR due to presenting the CS alone

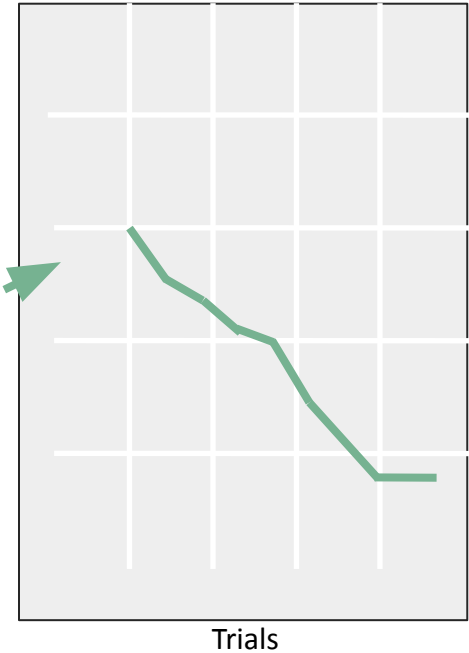


Is extinction just forgetting?

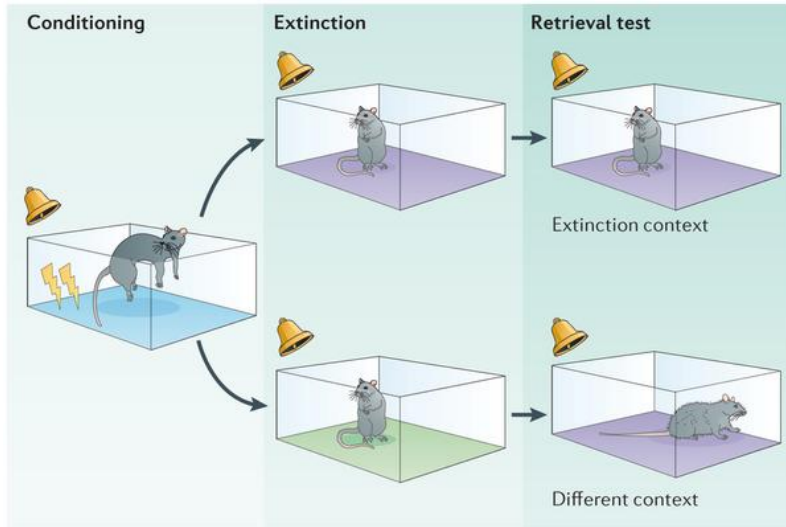
Spontaneous Recovery



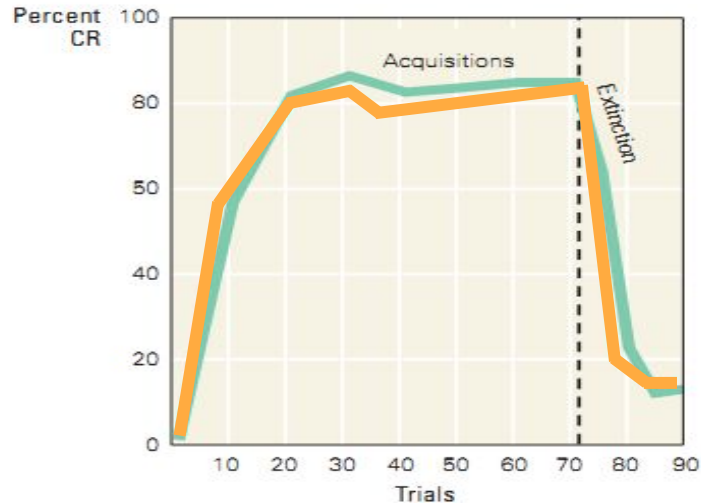
...
(time)



Renewal



(Maren et al., 2013)



- CR returns or 'renews' in a context when it changes from the extinction context

Extinction

- The association between the CS and US isn't simply forgotten
 - *Memories are not deleted*
- Rather, animals *learn something new*, which is that the CS does **not** predict the US
 - This knowledge competes with and *suppresses* the initial knowledge

Habituation vs. Extinction

● Habituation

- Non-associative learning
- Involves repeatedly presenting a stimulus to which you have an **automatic** response (e.g., loud noise causes you to startle)
- With repeated presentation, response decreases

● Extinction

- A phase of classical conditioning
- Occurs **after** the acquisition phase, and involves repeatedly presenting the CS **without** the US
- With repeated presentation, CR decreases

Brain Substrates of Classical Conditioning

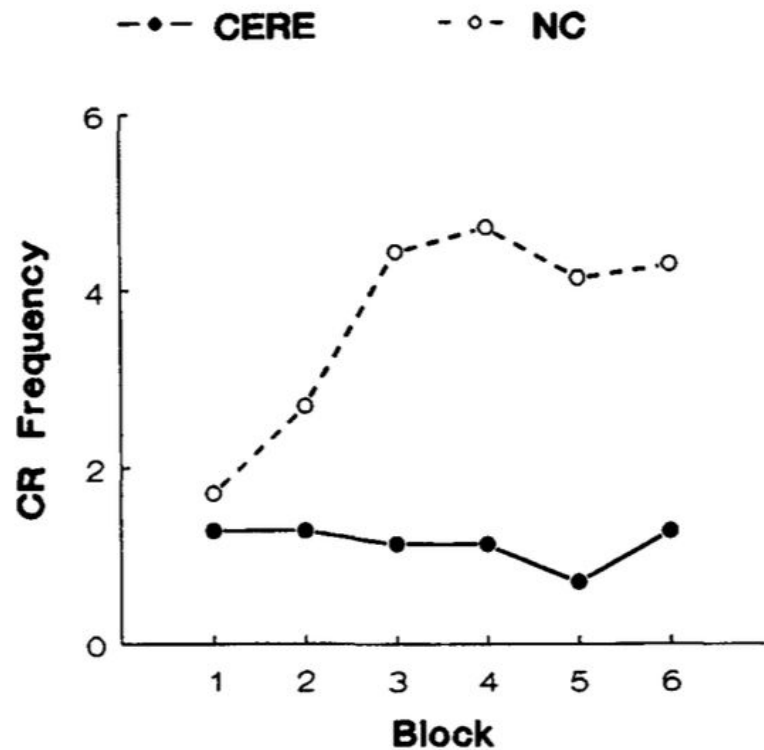
- There are many different types of classical conditioning that rely on different areas of the brain, depending on:
 - **Type of CS** (visual, auditory, taste, etc.)
 - **Type of US** (food, mate, pain, sickness, etc)
 - **Type of UR/CR** (salivating, freezing, blinking, etc)
- Two most well-studied brain substrates are for fear learning and eye-blink conditioning

Eye-blink Conditioning

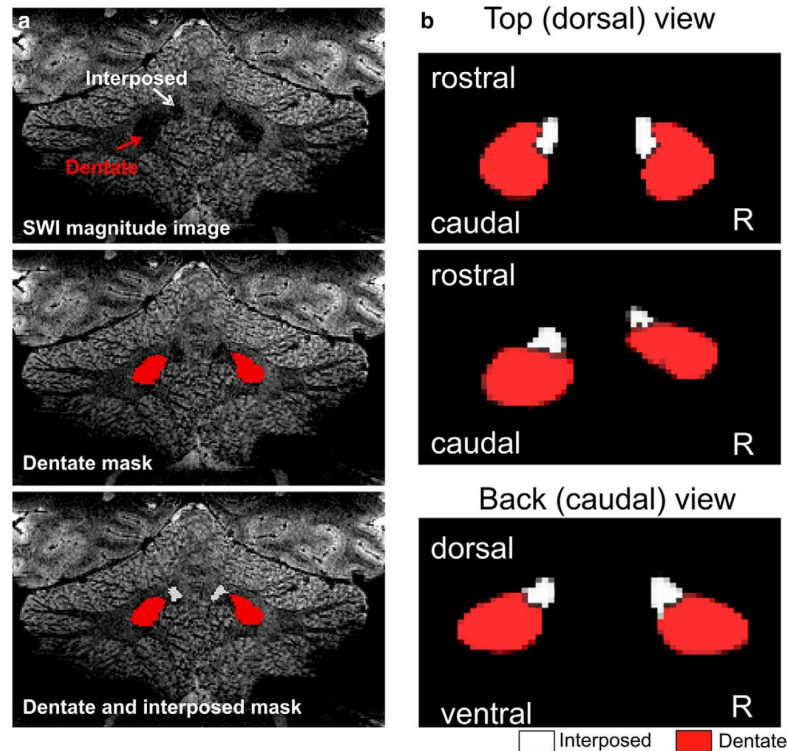
- Human and animal research both suggest that eye-blink conditioning is dependent upon the **cerebellum**



Human Eye-blink Conditioning



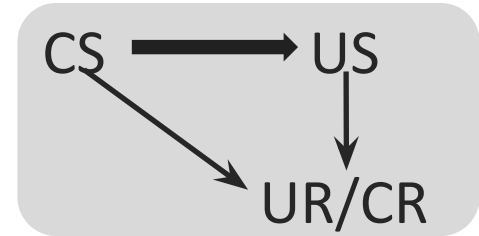
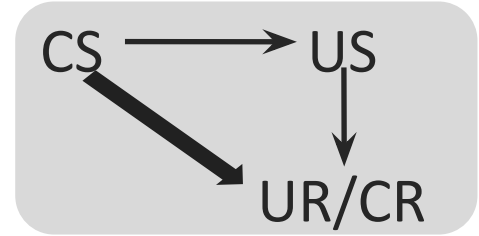
(Baum et al., 1993)



(Thürling et al., 2015)

What is learned in classical conditioning?

- Behaviorist view: A new S-R association
 - Learn a **new reflexive behavior**
- Cognitive view: A new S-S association
 - Learn to use the CS to predict the US
- How can we test these competing views?
 - US *devaluation* or *inflation*



Devaluation

	Acquisition	Devaluation	Test
Group 1	Light  Loud noise	—	Light ? <i>Strong fear</i>
Group 2	Light  Loud noise	Loud noise	Light ? <i>Low fear</i>

- S-R prediction
 - Animals learn a direct association between the light (CS) and fear (UR), so devaluation of the noise (US) will have no effect on fear
- S-S prediction 
 - Animals learn an association between the light (CS) and noise (US), so devaluing the noise (US) will decrease fear

Inflation

	Acquisition	Inflation	Test
Group 1	Tone → Mild shock	—	Tone ? <i>Mild fear</i>
Group 2	Tone → Mild shock	Strong shock	Tone ? <i>Strong fear</i>

- S-R prediction
 - Animals learn a direct association between the tone (CS) and fear (UR), so inflation of the shock (US) will have no effect on fear
- S-S prediction ✓
 - Animals learn an association between the tone (CS) and shock (US), so inflating the shock (US) will increase fear

Can anything be used as a CS?

- Rate of learning is determined by the extent to which the CS is *naturally relevant* to the US



CS

Sight/sound of wolf



Mushroom flavor



US

Leg injury



Stomach ache

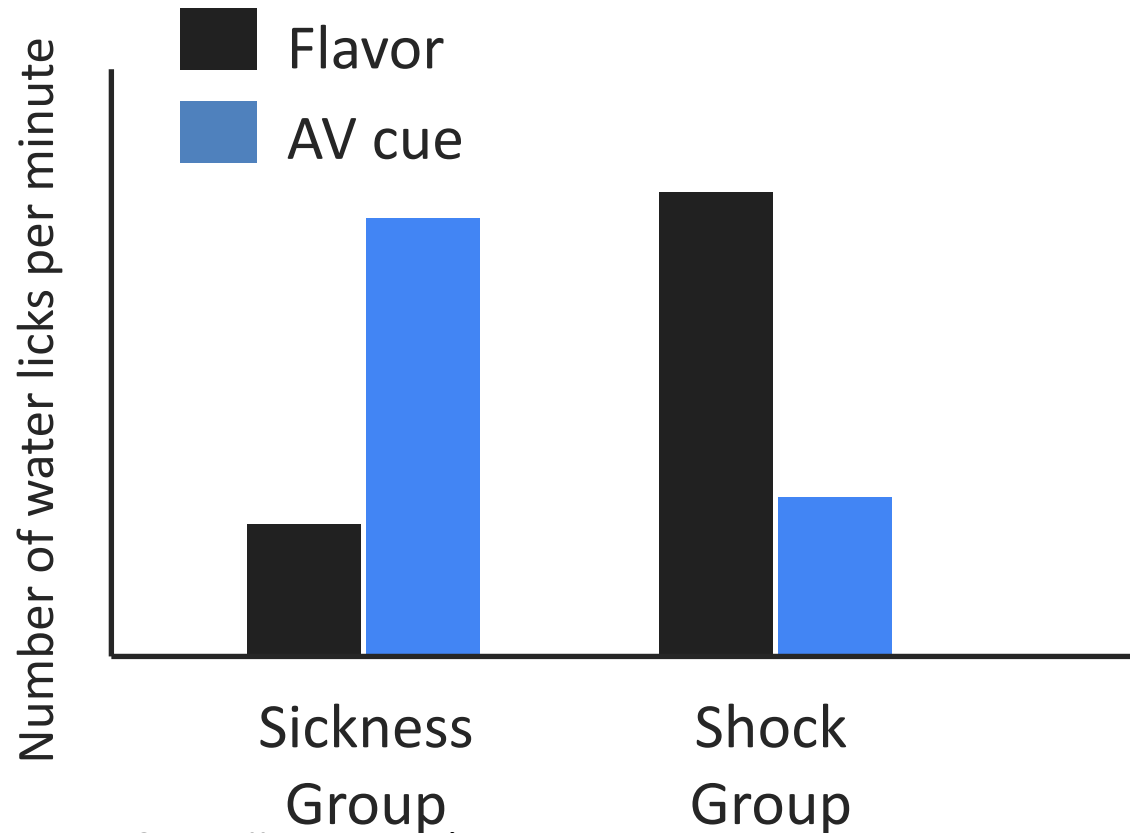


Conditioned Taste Aversion

Garcia and Koelling (1966)

- Conditioned two groups of rats with a compound CS (flavored water + audiovisual cue) that was administered when rat licked a spout
- US was either sickness or shock
- Tested response to each element of the CS independently (flavored water only OR audiovisual cue only)

Conditioned Taste Aversion



The Sickness group
'blamed' the flavored
water (less licking
when the flavor was
presented again)

The Shock group
'blamed' the AV cue
(less licking when the
AV cue was presented
again)

Outline

- Classical Conditioning Overview
- Rescorla Wagner Model
- Operant Conditioning

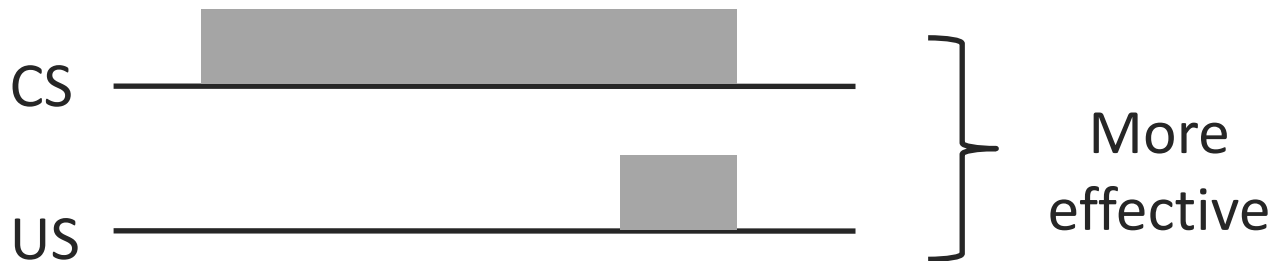
What causes conditioning?

Contiguity theory: Learning occurs by pairing the CS and US together in time

Contingency theory: Learning occurs when CS **predicts** the US

CS/US Timing

- Delay conditioning



- Trace conditioning



Challenges to Contiguity Theory

- Conditioned taste aversion
 - We can learn to avoid flavors (CS) associated with sickness (US) even when the two occur several hours apart
- Pairing the CS/US close together in time is *not necessary* for conditioning



Challenges to Contiguity Theory:

Blocking

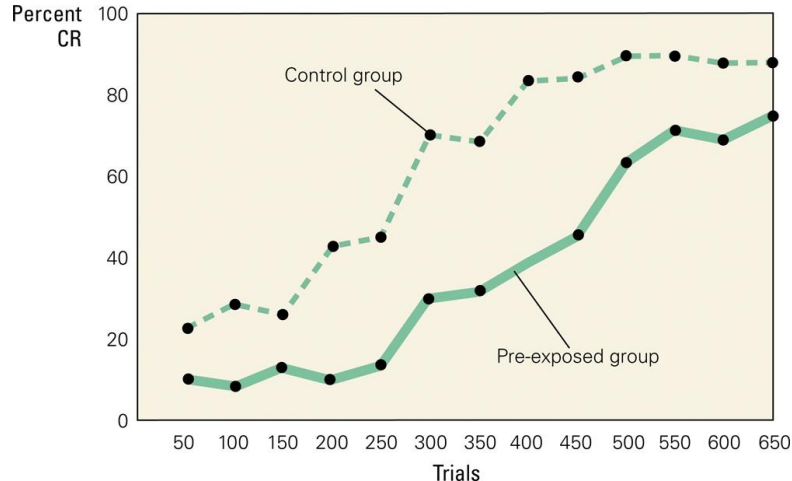
	Pre-condition	Conditioning	Test
Control	—	Light + Tone → Shock	Tone ? <i>Strong fear</i>
Pre-cond	Light → Shock	Light + Tone → Shock	Tone ? <i>No fear</i>

- Learning that the light predicts shock **blocks** the pre-conditioned group from learning about the tone
- Pairing the CS/US close together in time is *not sufficient* for conditioning

Challenges to Contiguity Theory: Latent Inhibition

Control	Pre-exposure —	Conditioning Tone → Shock
Pre-exp	Tone	Tone → Shock

- Pre-exposure to the tone inhibits this group from learning that the tone predicts shock
 - Called **latent inhibition**
- Pairing the CS/US close together in time is *not sufficient* for conditioning



Contingency Theory

- Learning occurs if the CS *predicts* the US
 - The amount of conditioning is proportional to the amount of information that the CS provides
- Helps explain blocking:
 - Light (CS₁) already fully predicts shock, so animals don't learn about the tone (CS₂)
- The main proponent of this view, Robert Rescorla, developed a formal mathematical model of learning with Allan Wagner

Rescorla



Wagner



Rescorla-Wagner Model

- The amount of learning that takes place on a given trial is related to how surprising the US is
 - Amount learned = US received – US expected
- Formal model: $\Delta v = k (\lambda - v)$

Δv	=	k	(	λ	—	v	)
Amount learned		Learning rate parameter		US Received		US Expected	
		↓				↓	
		Related to CS salience				Will determine the CR	

Rescorla-Wagner Model

- Using the classic Pavlov example:
 - On each trial, play a mildly salient tone ($k = 0.5$) and give 1 piece of food
- Plug in these values for the first trial:

$$\underset{\substack{\mathbf{V} \\ \text{Amount learned}}}{\Delta} = \underset{\substack{\text{Learning rate} \\ \text{parameter}}}{0.5} \left(\underset{\text{US received}}{1} - \underset{\text{US expected}}{0} \right) = 0.5$$

- New expectation = initial expectation + what was learned

Rescorla-Wagner Model

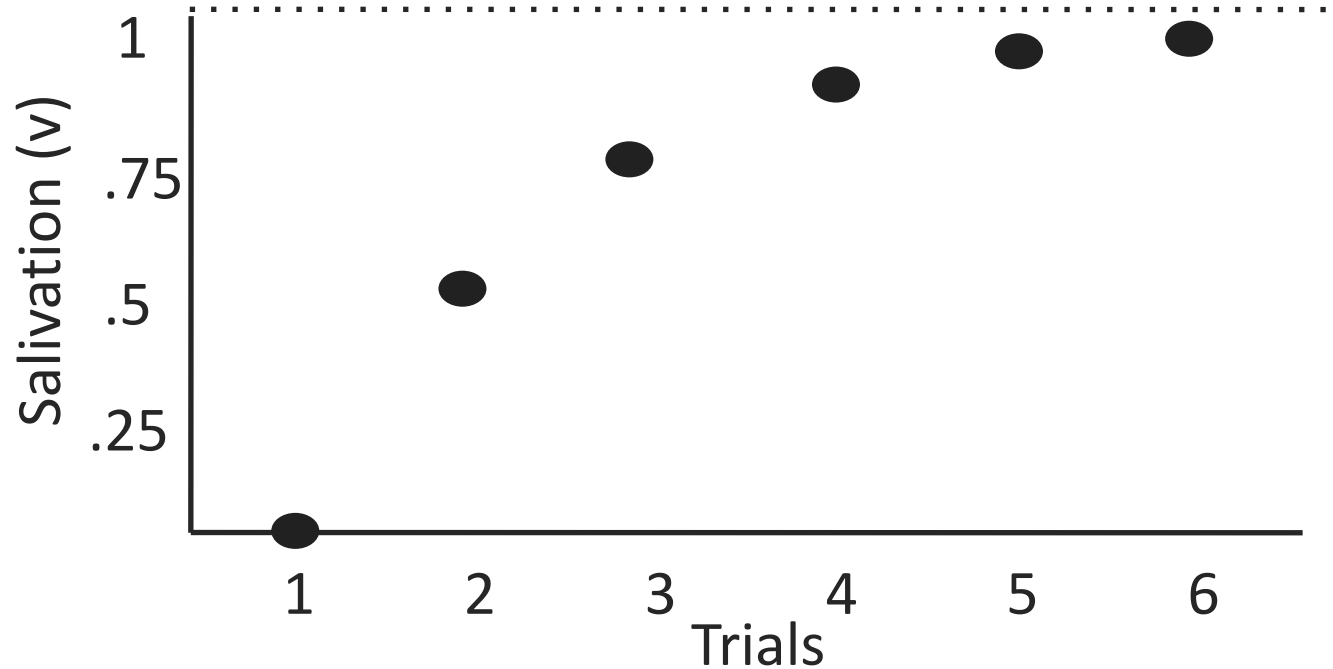
- Trial 2

$$\underset{\substack{\mathbf{V} \\ \text{Amount learned}}}{\Delta} = \underset{\substack{\text{Learning rate} \\ \text{parameter}}}{0.5} \left(\underset{\text{US received}}{1} - \underset{\text{US expected}}{0.5} \right) = 0.25$$

- Trial 3

$$\underset{\substack{\mathbf{V} \\ \text{Amount learned}}}{\Delta} = \underset{\substack{\text{Learning rate} \\ \text{parameter}}}{0.5} \left(\underset{\text{US received}}{1} - \underset{\text{US expected}}{0.75} \right) = 0.125$$

Rescorla-Wagner Model



Rescorla-Wagner Model

- This very simple model can explain:
 - Acquisition (surprised to receive US) ✓
 - Extinction (surprised *not* to receive US) ✓
 - Blocking (*not* surprised to receive US because CS_1 predicts it) ✓
- Cannot, however, explain *latent inhibition*
 - During pre-exposure to the tone there is no *prediction error*
 - Thus the model predicts identical performance for both groups ✗

Outline

- Classical Conditioning Overview
- Rescorla Wagner Model
- Operant Conditioning

Operant Conditioning



Jon-Michael Sullivan/ZUMA
Press/Newscom

Operant Conditioning

- Pioneered by Edward Thorndike and B.F. Skinner
- Mechanism by which animals learn an association between a response and its outcome (R→O)
 - *Example: Child learns that if it cries, it will be held*
- Operant conditioning can serve to either increase or decrease a behavior based on reinforcement

How is operant conditioning different from classical conditioning?

Types of Operant Conditioning

	Reinforcement Increases behavior	Punishment Decreases behavior
Positive Consequence added	Giving a rewarding stimulus	Giving an aversive stimulus
Negative Consequence removed	Removing an aversive stimulus (Escape)	Removing a rewarding stimulus (Omission)

Types of Operant Conditioning

	Reinforcement Increases behavior	Punishment Decreases behavior
Positive Consequence added	Dog shakes hands; receives treat	Cat scratches sofa; is sprayed with water
Negative Consequence removed	Put on seatbelt; beeping stops	Child hits sibling; toy is taken away

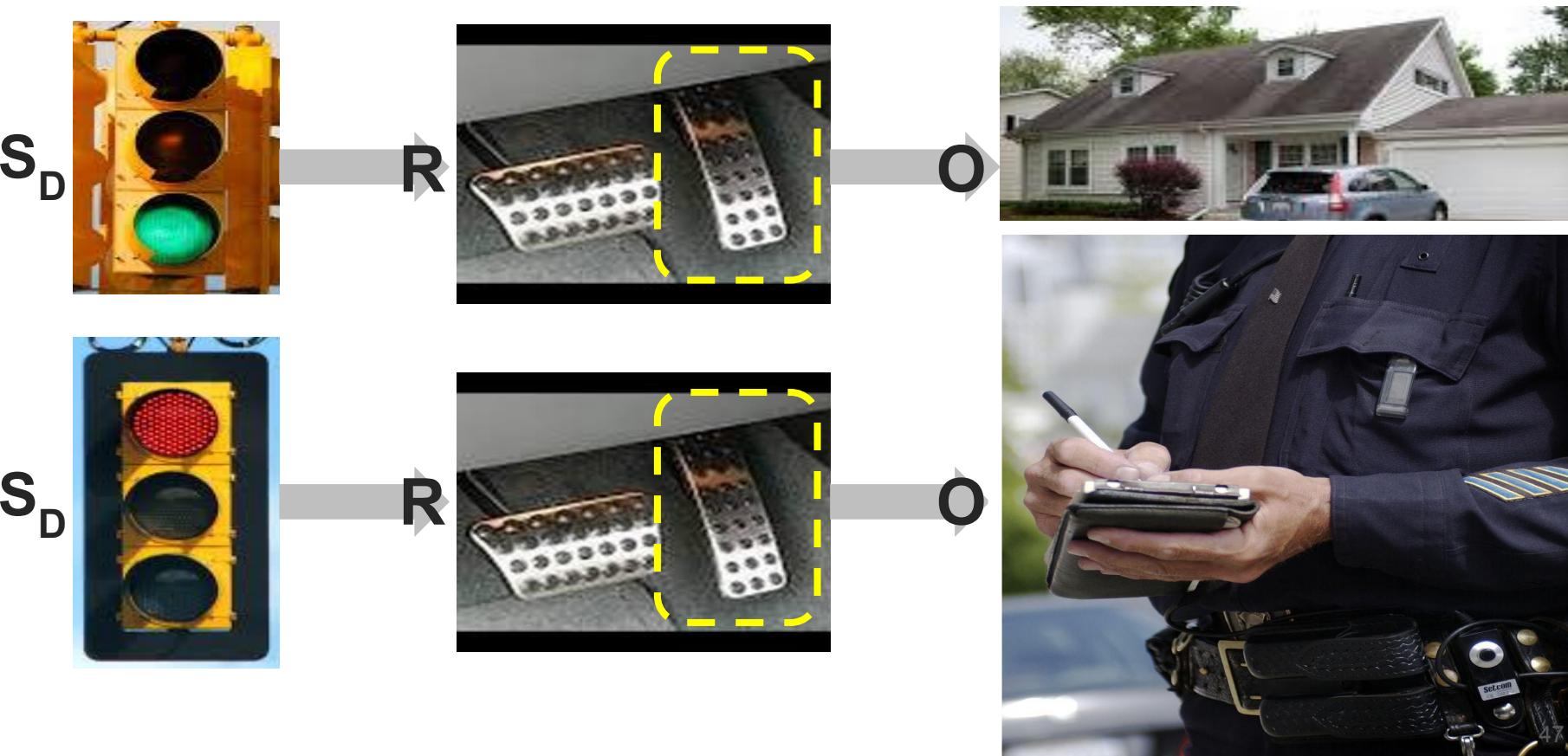
What is the “R” in $R \rightarrow O$?

- Initially thought to be a very specific series of movements
- We now know that learning is more general
 - Rats trained to lever-press for food with their paw will use their nose if the paws are immobilized
 - Rats trained to run through a maze to obtain food will also swim through the same maze
- “R” is not a specific motor response, but rather a class of behaviors that brings about an outcome

The Role of Context

- Animals also learn that $R \rightarrow O$ associations are often only true in certain *contexts*
- Operant conditioning more broadly includes learning about $S_D \rightarrow R \rightarrow O$ relationships
 - **Discriminative stimulus (S_D)**: ensures that we select behaviors that are appropriate for the situation (“context”)

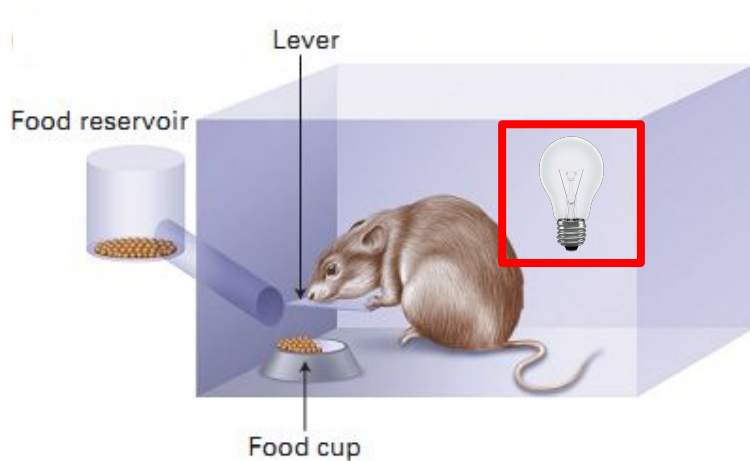
The Role of Context



Free-operant Paradigms

- Animal operates the testing apparatus freely, so that it controls the rate of responding

Skinner Box



- Response (R): lever press
- Outcome (O): food delivery
- Discriminative stimulus (S_D): light

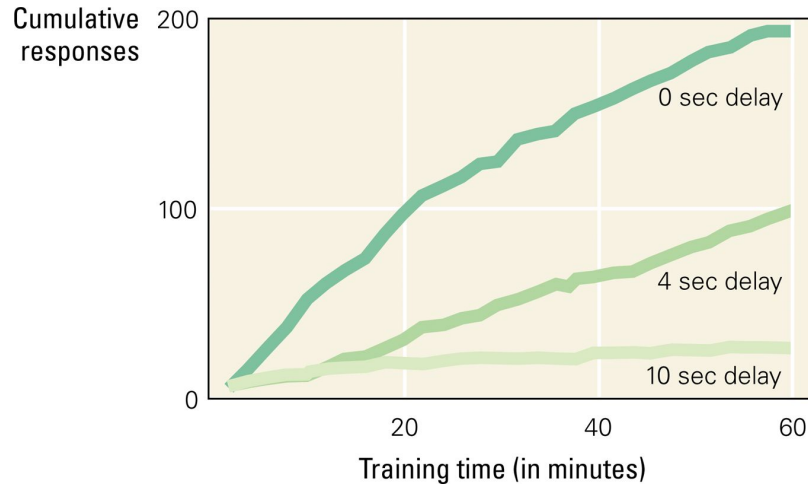
Shaping

- **Shaping**: reinforcing *successive approximations* of the desired behavior
 - Example: Lever-pressing in rats
 - Approach the lever... touch the lever... rear up and touch the lever... press down on the lever
- **Chaining**: training animals to execute not just a single complex behavior, but a *sequence* of behaviors



Immediacy

- The closer in time the response and the outcome, the more rapid the learning

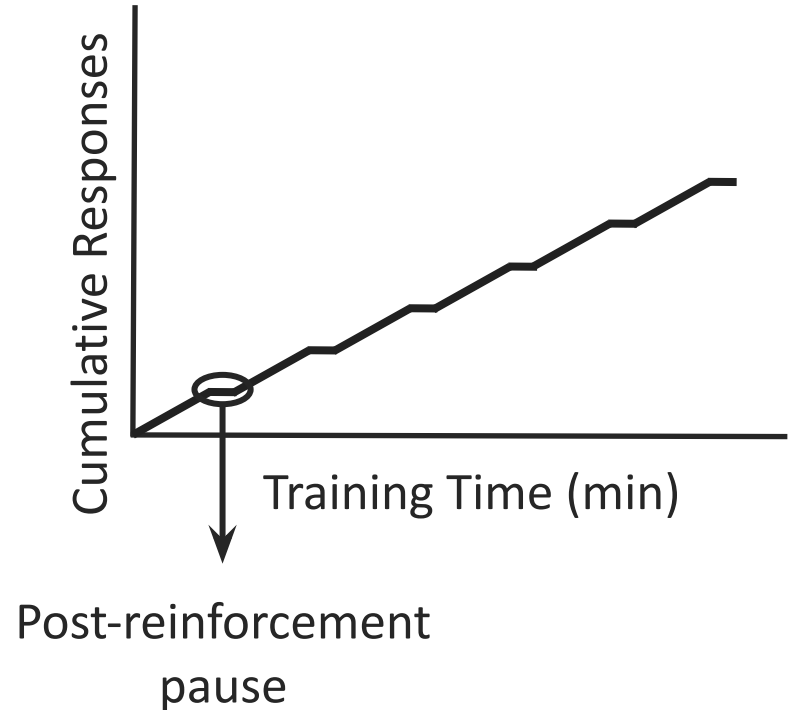


Contingency

- The $R \rightarrow O$ contingency, i.e., how frequently a behavior is reinforced/punished, significantly affects learning
- **Continuous** reinforcement schedule
 - Each response is followed by an outcome
- **Partial** reinforcement schedules
 - Responses are intermittently followed by outcomes
 - More naturalistic scenario

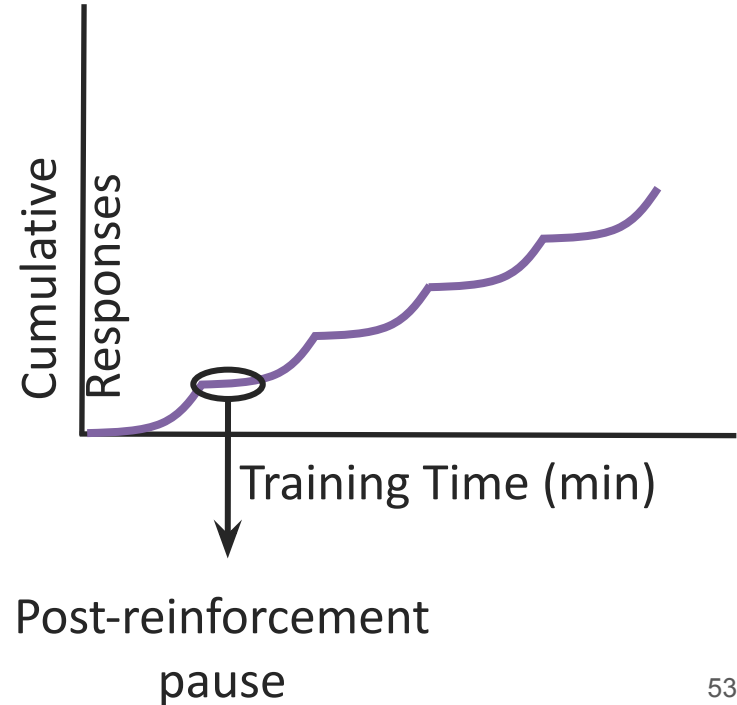
Fixed Ratio Schedule (FR)

- X number of responses must be made before reinforcement
- For example:
 - FR 3 = reinforcement follows every 3rd response
 - FR 10 = reinforcement follows every 10th response



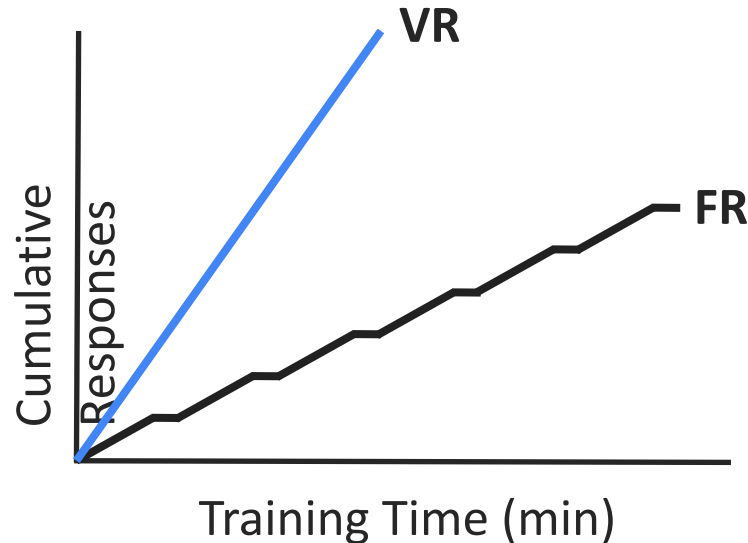
Fixed Interval Schedule (FI)

- Animals are reinforced for the first response made after X amount of time
 - E.g., FI 30s = reinforced for the first response made 30s after the last reinforcement
- “Scallop” pattern results from animals not being able to precisely judge time



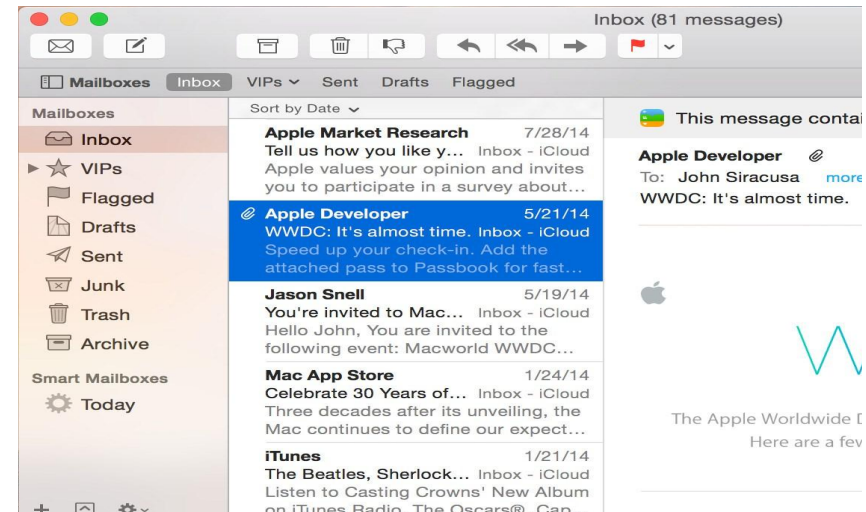
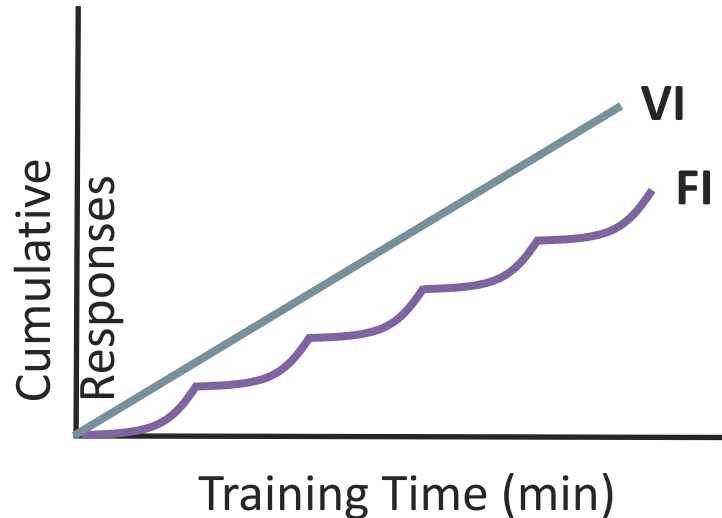
Variable Ratio Schedule

- *On average*, X number of responses must be made before reinforcement
- Encourages a rapid and more constant rate of responding than FR schedules



Variable Interval Schedule (VI)

- *On average*, animals are reinforced for the first response made after X amount of time
 - Encourages a moderately rapid, steady rate of responding



Identify the Reinforcement Schedule

1. A first-grade teacher gives each student a gold star if the child completes that day's math worksheet. Five gold stars can be exchanged for a prize.

Fixed-Ratio

2. A good telemarketer makes an average of two sales for every 20 phone calls he makes, so he earns the most profit if he makes lots of calls.

Variable Ratio

3. A couple go to their favorite restaurant on a Saturday night and are told that a table will be available in about 30 minutes. They wait in the bar and periodically return to check whether a table is free.

Variable Interval

4. Maria donates blood regularly at the local hospital. Due to hospital policy, donors have to wait 2 weeks between donations.

Fixed Interval

Concurrent Schedules

- Animal can make two or more responses, each leading to different outcomes
 - E.g., press lever A = fruity pebbles, press lever B = cocoa krispies
- **Matching law**: response rates are proportional to reinforcement rates

$$\frac{\text{Rate of responding A}}{\text{Rate of responding B}} = \frac{\text{Rate of reinforcing A}}{\text{Rate of reinforcing B}}$$

- If pressing lever A is reinforced 5x more often than pressing B, animals will press A 5x more often than B

Punishment

- Reinforcement is not the only way to influence behavior!
- **Punishment** can reduce unwanted behaviors, but using it effectively can be tricky.
- Why might punishment be less effective than rewards?
 - Does not indicate desired behavior
 - Circumvention
 - Simultaneous reinforcement

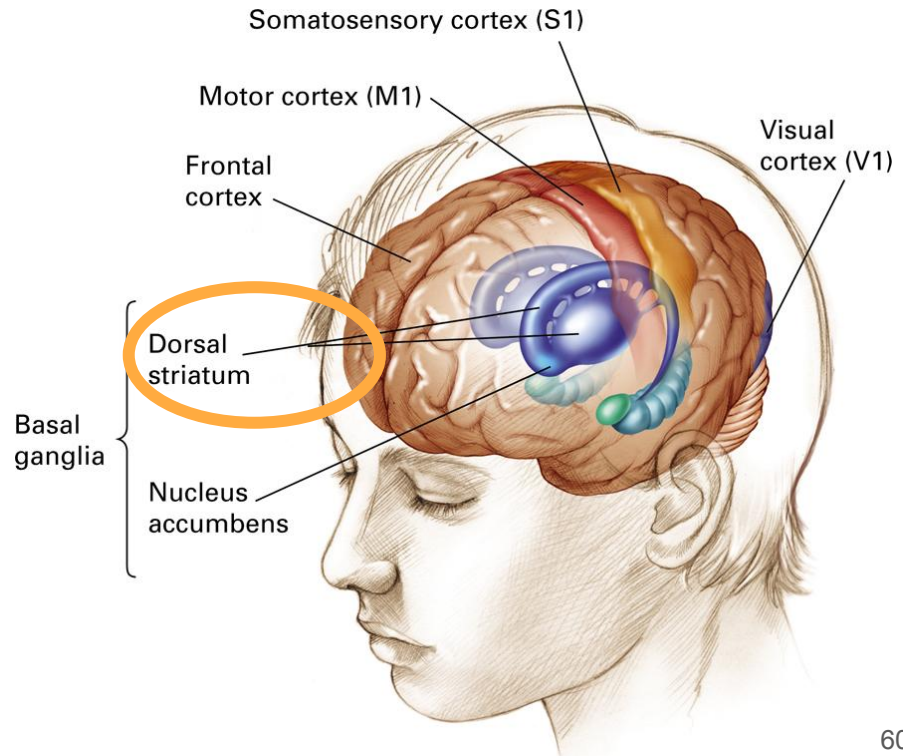


Punishment

- How can punishment be used effectively?
 - Start with a strong punishment
 - Punish immediately
 - Punish consistently
 - Reinforce alternative, desired behaviors

Neural Basis of Operant Conditioning: Dorsal Striatum

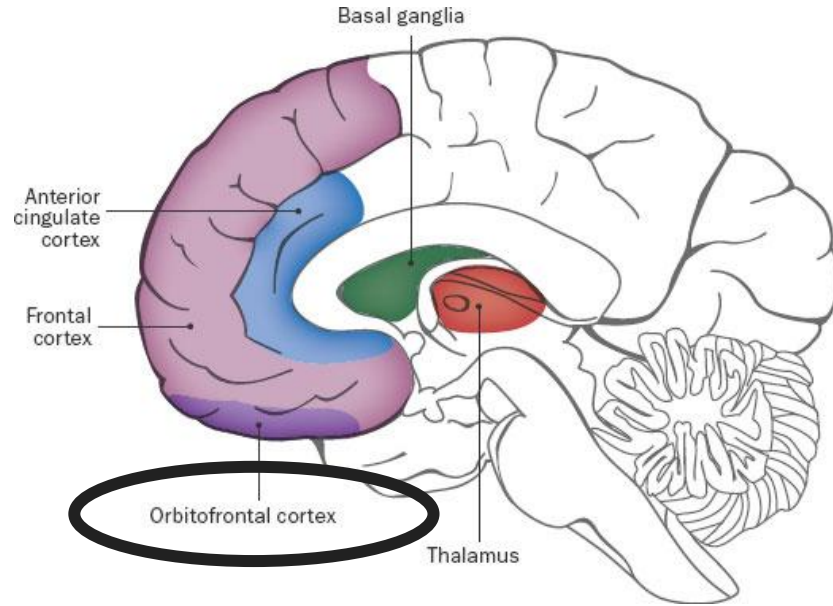
- Part of the basal ganglia, comprised of the caudate and the putamen.
- Receives input from sensory areas and sends projections to motor areas



Dorsal Striatum

- What effects do lesions to the dorsal striatum have on operant conditioning?
- Damage to the dorsal striatum:
 - Leaves intact the ability to learn simple **R**→**O** relationships
 - Impairs the ability to learn about **discriminative stimuli** ($S_D \rightarrow R$ associations)
 - Reveals that this region is necessary for learning which **responses** (R) are appropriate in which **contexts** (S_D)

Neural Basis of Operant Conditioning: Orbitofrontal Cortex



OFC Damage

- Leaves intact the ability to learn about discriminative stimuli ($S_D \rightarrow R$ associations), but...
- Difficulty learning if outcomes are switched

Red light $\rightarrow$ Press lever $\rightarrow$ Food

Blue light $\rightarrow$ Press lever $\rightarrow$ No food

- OFC is important for **reversal learning of $R \rightarrow O$ associations**: Updating responses (R) to be associated particular outcomes (O)

Next Time

- Cognitive Control